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Safe and Sanitary Report

FOR 6C Unit 4 Churton Street. Parnell

Auckland

Prepared for

Kent and Lauren McIntyre

Job Number

CHUR 6

Date

15th Dec 2022

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BACKGROUND:

This report has been prepared by Archavium Ltd at the instruction of the owners for the property located at 6C unit 4 Churton Street, Parnell for works undertaken circa 1985-1990 encapsulating the infilling of a permitted garage to that of a habitable bedroom.

The house inspection and report was undertaken by Vince May, Vince is experienced in this kind of report having spent over 35 years in the industry as an architectural designer for the first 20 years and then as remediation consultant and lecturer in building science for 10 years.

Vince has been authoring COA, safe and sanitary and durability reports that have been submitted and accepted by Auckland Council over the last 8 years in conjunction with Archavium Ltd's design practice.

The dwelling at unit 4, 6C Churton Street is one of four attached townhouses of approximately 158m². The home was originally constructed in 1982 Consisting of engineered design concrete block intertenancy and lower floor walls and light weight timber framing for the remainder.

The legal description is:

Lot 1, DP 98722

CONSTRUCTION SEQUANCE:

The original building permit issued in 1982 for the construction of a two-bedroom two-story townhouse with a lock up garage

A second building permit was applied for prior to 1st July 1992 and was granted around December 1992. It is to be understood by the limited documentation that as it was processed by Auckland Council pre the 1991 Building Act and as such the building permit HB92/2496 was issued as opposed to a building Consent. This included the encapsulating of the existing upper-level deck into a permanent sunroom. With the addition of a garage/storeroom being constructed and encapsulated beneath.

Either at the time of the original building works or most likely fairly soon thereafter, the permitted garage was converted to a bedroom. This has been deducted (as there is no documented evidence) based on the following factors.

The aluminium joinery in the converted bedroom 3 matches the existing house joinery and has the same architrave detail as the existing property as opposed to the more modern grooved jamb liners present in the sunroom extension of 1992.

All finishing elements inside the converted bedroom match the existing home.

For as far back as real estate sales records show the dwelling has been marketed as 3 bedrooms.

Our best estimate is that it was converted not long after the initial build as the built environment still shows a clear indication of the garage door opening being built and remaining in place.

For the purpose of this report, it can be taken that the conversion was undertaken prior to the building Act of 1991 taking effect (between the 1982 and 1992 building permits) and as such this safe and sanitary report will be required be placed on Council file.

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This safe and sanitary report is limited to the conversion of the lower floor garage into a habitable living space.

Where a safety or compliance issue was noted by the author, recommendations have been made with regards to other facets of the built environment (outside of the above scope). This is not intended as a full building survey report. See the suggestions section.

Although many facets of the current New Zealand building code were not applicable at the time this building was constructed. It is the authors opinion that the current building code is a good measure for proving that a building is safe and sanitary for use.

The report below compares the un-permitted works to the current performance clauses in the NZBC.

BUILDING REPORT:

1. B1 – Structure

NZBC B1 – Structure, performance clauses B1.3.1 – B1.3.4 must be considered when assessing existing built structures for compliance.

> *B1.3.1 Buildings, building elements and sitework shall have a low probability of rupturing, becoming unstable, losing equilibrium, or collapsing during construction or alteration and throughout their lives*
>
> *B1.3.2 Buildings, building elements and sitework shall have a low probability of causing loss of amenity through undue deformation, vibratory response, degradation, or other physical characteristics throughout their lives, or during construction or alteration when the building is in use.*
>
> *B1.3.3 Account shall be taken of all physical conditions likely to affect the stability of buildings, building elements and sitework, including:*
>
> *a) Self weight*
>
> *b) Imposed gravity loads arising from use*
>
> *c) Temperature*
>
> *d) Earth pressure*
>
> *e) Water and other liquids.*
>
> *f) Earthquake*
>
> *g) Snow*
>
> *h) Wind*
>
> *i) Fire*
>
> *j) Impact*
>
> *k) Explosion*
>
> *l) Reversing or fluctuating effect*

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m) Differential movement

n) Vegetation

o) Adverse effects due to insufficient separation from other buildings

p) Influence of equipment, services, non-structural elements and contents

q) Time dependent effects including creep and shrinkage

r) Removal of support

Whenever possible visible elements were inspected, and the findings are below.

1.1 Structure of walls floor and roof for converted bedroom 3.

1.1.1. **B1.3.1** The wall, floor and roof structures were built in accordance with the plans supplied in the 1982 building permit documentation.
These works were designed by a structural engineer and inspected and approved during the building inspections undertaken in 2003

The structure has been in place since 1982-3 and during visual inspections there were no signs of the structural integrity of the building being compromised by either sagging, fracturing or movement. This visual inspection is one of the indicators used to prove that a structure has performed in accordance with NZBC B1 structure for in excess of 35 years or 70% of its minimum intended performance life and is showing no current signs of failure.

1.1.2. **B1.3.2** During visual inspection no signs of deformation and/or degradation of the structure were observed. Based on the proof of compliance demonstrated in 1.1.1 B1.3.1 above, we can conclude the structure is compliant with the performance clause B1.3.2 of the NZBC.

1.1.3. **B1.3.3** According to NZBC B1.3.3, account shall be taken of all physical conditions likely to affect the stability of the building, building elements and siteworks. The conditions are:

a) **Self weight.** The structure was built as per the consented permit applications plans and specifications and consist of engineered concrete block walls, concrete floor slab and light timber framing for the midfloor above. There are no detectable signs showing that the structure has not being capable of supporting its self-weight or any reason as to why it will not continue to do so.

b) **Imposed gravity loads arising from use.** As above there has been no damage to the structure that is visible from use over the last 35 years

c) **Temperature**. There has been no damage visible from use over the last 35 years

d) **Earth pressure**. There has been no damage visible from use over the last 35 years.

e) **Water and other liquids.** There is no signs of internal moisture issues caused by an ingress of moisture to the structure either through the exterior envelope, the foundations or from internal moisture in bedroom 3 so it can be assumed that the concrete block walls and associated protective coatings have performed in accordance with the building code and have protected the structure from moisture. It is also noted that the structure is reinforced concrete block so as a

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durable material this is unlikely to deteriorate with small quantities of moisture and other parts of the structure is like for like and shows no deterioration.

f) **Earthquake.** The building was compliant with the code or the day. Since the time of initial construction circa 1983 there is no sign of any structural fracturing due to lateral forces.

g) **Snow.** No snow loads are considered in the Auckland region

h) **Wind.** See f) Earthquake

i) **Fire.** No fire protection is required to the un-permitted third bedroom, as it is encompassed entirely within the single house dwelling fire cell.

j) **Impact.** N/A

k) **Explosion.** No explosive devices are installed.

l) **Reversing or fluctuating effects.** Structure is proven to be compliant with the building code

m) **Differential movement.** Structure is proven to be compliant with building code.

n) **Vegetation.** The structure is free of any vegetation

o) **Adverse effects due to insufficient separation from other buildings.** There are no adverse effects from proximity to other buildings.

p) **Influence of equipment, services, non-structural elements and contents.** N/A

q) **Time dependent effects including creep and shrinkage.** No such effects were found after 35 years.

r) **Removal of support.** No supports have been removed.

All structural elements were permitted in 1982.

2. B2 – Durability

NZBC B2 – Durability, performance clauses B2.3.1 a) - c) must be considered when assessing existing built structures for compliance. As the practical completion of the structures was in early 1983 all elements with a 5 and 15 year durability requirement have already passed their minimum durability requirement set out in NZBC B2 – Durability. And all elements requiring a no less than 50 year durability requirement are 70% of their minimum durability performance requirement (2033).

NZBC B2 -Durability places building elements into a performance requirement of 1 of 4 categories.

1. Nil – for those elements that do not have a performance requirement under the NZBC. (passed in all dwellings)

2. 5 yrs for any elements where: (durability requirement from initial elements was passed and therefore performance complied with in 1988)
    a. The building elements (including services, linings, renewable protective coatings, and fixtures) are easy to access and replace, and

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b. Failure of those building elements to comply with the building code would be easily detected during normal use of the building.

3. 15 years if: (durability requirement from initial elements was passed and therefore performance complied with in 1998)
   a. Those building elements (including the building envelope, exposed plumbing in the subfloor space, and in-built chimneys and flues) are moderately difficult to access or replace, or
   b. Failure of those building elements to comply with the building code would go undetected during normal use of the building but would be easily detected during normal maintenance.

4. The life of the building, being not less than 50 years, if: (2033)

   Those building elements (including floors, walls, and fixings) provide structural stability to the building, or

   a. Those building elements are difficult to access or replace, or
   b. Failure of those building elements to comply with the building code would go undetected during both normal use and maintenance of the building.

2.1 Structural

The materials in the existing structural elements, consist of reinforced concrete or masonry and Radiata pine and their associated fixings.

There are no visible signs of deterioration with regards to surface cracking of gypsum plaster walls or sagging within the building elements.

The materials used for structural elements were permitted in 1982 as part of the initial building consent. Due to the nature of the original build and the length of time the structure has been in place there is substantive proof under the performance criteria of the NZBC that the building will continue to perform in accordance with NZBC B2-Durability with regards to structural elements.

2.2 Building envelope

The envelope is a combination of a 'mass', non-cavity rain screen and aluminium joinery.

- The 'mass' portion of the envelope consists of a concrete block wall with acrylic breathable paint. The concrete block wall has a minimum durability requirement of 50 years and concrete block at the time of construction and in accordance with current acceptable solutions has a durability of greater than 50 years so is compliant. The acrylic paint membrane is a breathable paint on membrane that has a durability requirement of no less than 5 years. The acrylic paint is in a well-maintained state and at the time of construction and in accordance with current acceptable solutions has a durability of greater than 5 years so is compliant with the NZBC.

- No destructive testing of the rusticated timber weatherboards to the infill panel where the garage door was initially in place was undertaken. Rusticated weatherboards in this environment were and

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are now an acceptable material for the durability requirement of 15 years as described in the current acceptable solutions. It is very likely that the weatherboards are either Cedar or radiata pine, and these have been protected from the elements with a well maintained acrylic paint system showing no signs of decay, so have met the minimum requirements of the NZBC of 15 years and provided the paint system is continued to be maintained in accordance with NZBC B2/Durability, will continue to perform for the foreseeable future.

- Aluminium joinery has a minimum durabilty requirement of 15years and these windows have been performing for over 30 years without any visible signs of moisture ingress. As such they have performed in accordance NZBC B2 – Durability, and with regular maintenance will continue to do so.

The exterior envelope recently maintained and is complying with the durability requirements of the NZBC.

3. D1 – Access

NA to this report

4. E1 – Surface water

5. NA to this report

6. E2 – External Moisture

The relevant performance clause's that would apply for this project would be:

E2.3.2 Roofs and exterior walls must prevent the penetration of water that could cause undue dampness, damage to building elements, or both.

E2.3.5 Concealed spaces and cavities in buildings must be constructed in a way that prevents external moisture being accumulated or transferred and causing condensation, fungal growth, or the degradation of building elements.

E2.3.2

Roof claddings are not part of this report and have not been inspected.

Exterior wall claddings from the 1982 building consent consist of mass construction (reinforced concrete block), with an acrylic paint finish. This was an accepted cladding technique utilised in this era of construction and is currently permitted for use with on this dwelling on a habitable space being bedroom 2 (adjacent bedroom 3). See image below.

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Mass construction for exterior cladding is described by the DBH document "External moisture – An introduction to weathertightness design principles as; '*Mass walls rely on thickness to provide sufficient absorption in the outer layers without affecting inner surfaces. They also require sufficient durability to accommodate continual wetting and drying without deterioration of the material. Mass walls do not respond well to external weather sealing or coatings restricting vapour transference. Interruption of 'breathability' of the wall material can lead to moisture build-up within the walls'*.

The concrete block wall by itself is not dense or thick enough to perform in NZ as a 'mass wall' so the accepted method (as was done here) is to paint the wall with an acrylic paint system to provide weathertightness. And prevent the moisture from penetrating the concrete block whilst allowing moisture that does gain access either from an external or internal source to vaporise and pass through the acrylic coating when subjected to sunlight and warmth.

The junction of the Concrete block wall to the impermeable paving on the exterior driveway does not have adequate ground clearance to comply with the current acceptable solution NZBC E2/AS1, however this was not the code at the time and the methodology used was common practice.

In these situations, it is best to determine if the junction is performing by applying the performance criteria of the current NZBC. In this case has moisture been prevented from entering the structure to prevent *undue damp- ness, damage to building elements, or both.* The building has been maintained to such an extent that it is not possible to detect any moisture ingress damage or dampness to any internal building elements so under that criterion the wall is performing in accordance with NZBC E2:3.2.

Too better determine the effectiveness of compliance it is prudent to consider the methods go for which moisture can ingress a building. One of those is for moisture to build up or pond against the outside junction. In this case the driveway surface falls away from the building for on a continuous pathway so there is no chance that water can accumulate at the junction. It is therefore the opinion of the author that the exterior envelope has and will continue to perform as intended for the following reasons.

- There is no signs of failure at any of the junctions.

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- The envelope is built in accordance with construction practices at the time of construction.
- There are other junctions in the built environment that were permitted in this building and are still performing.

A common issue with infilling non-habitable spaces with habitable spaces is the admission of ground water through concrete masonry walls if they are not protected from groundwater with a suitable waterproof membrane. It has not been possible to determine if this membrane is in place, and destructive testing was not deemed appropriate or necessary. The retained portion of the new bedroom 3 is an integral part of the remainder of the structure where the structure is adjoining habitable rooms (and was in 1983). There are no signs of moisture ingress through any of the concrete block retaining walls in any area of the building so after 30+ years it is relatively safe to assume that there is moisture protection in place, or that the topography and layout of the other 3 townhouses effectively removes ground water from the back faces of the concrete block walls. In either case the foundation walls have successfully prevented moisture from entering any facet of the building over its 35 years of performance.

E2.3.5

The mass cladding system does not have any concealed spaces or cavities and the direct fixed rusticated weatherboard cladding system used appears to be in good condition. There is no obvious evidence of moisture penetration be found in the building – no mould growth or moisture accumulation was found during the inspection, and no damaged or moisture damaged skirtings or gypsum linings was evident.

7. E3 – Internal Moisture

N/A to this report

8. F2 – Hazardous Building Materials

N/A to this report

9. F4 – Safety from Falling

N/A to this report

10.G9 – Electrical

Electrical works in the existing building are deemed to comply with clauses G9.3.1 (a) – (g) and clause G9.3.3

11.G12 – Water Supplies

N/A to this report

12. G13 – Foul Water

N/A to this report

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13.H1 – Energy efficiency

Bedroom 3 is fully strapped and lined and as such was compliant with the built environment of the 1980's

Conclusion:

The works that were undertaken in the late 1980's were consistent with the building practices of the time.

Bedroom three has been used as a bedroom for the last 4-5 decades with many owners and families using the room as a bedroom. The performance of the bedroom has been well tested by time and as proven in the above report still comply with many of the current requirement of the NZBC with the exception of wall insulation.

Signed

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